

Project Zero-Watt:
AI-Driven Energy Optimization for 5G

Reducing 5G OpEx & Carbon Footprint via Predictive AI

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Executive Summary

Project Zero-Watt: Eliminating Idle 5G Energy Waste

Problem
Idle Capacity

5G towers consume 100% energy 24/7, even during zero-traffic windows.

Solution
Predictive "Deep Sleep"

XGBoost pipeline predicts traffic gaps 60-min ahead to trigger 'Deep Sleep'.

Impact
Multi-Million Dollar Savings

25% Energy Reduction,
$2.7M - $4.8M
OpEx Savings.

The Problem

5G Hardware Wastes ~90% of Power on Idle Traffic

- **5G towers run at 100%** power even when traffic is near zero.
- **Result:** Massive OpEx waste and unnecessary carbon emissions.
- **Potential savings:** Opportunity: $2.7M+ Annual OpEx Reduction (assuming 50k towers).



Solution: System Architecture

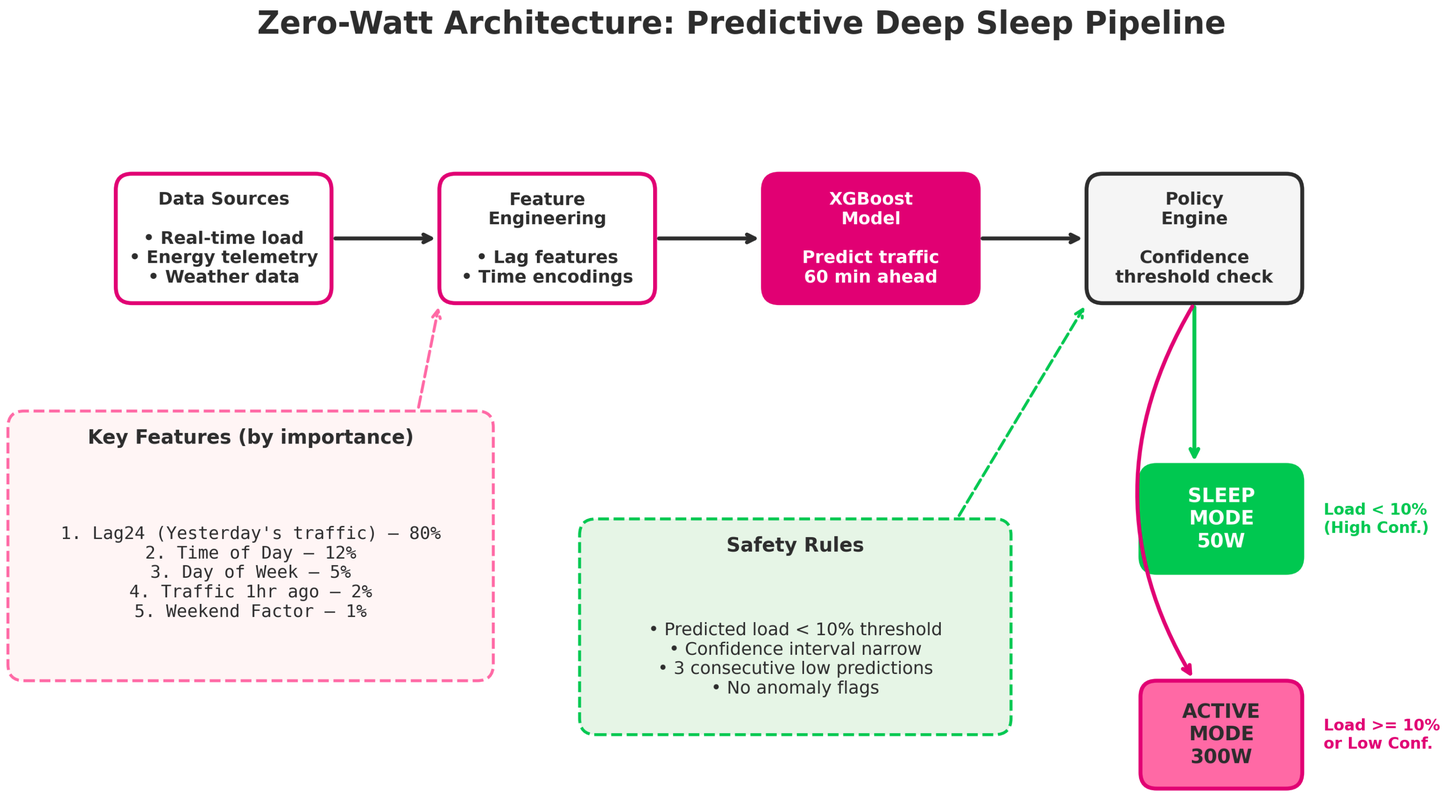
How Predictive "Deep Sleep" Protocol Works

2-Step Process:

1. **Inference (Regression):** Model predicts precise traffic load (e.g., 8.4%) 60-min ahead.
2. **Policy (Threshold):** Deterministic rule triggers "Sleep" only if Predicted Load < 10%.

Validation Metrics:

- **Accuracy:** Mean Absolute Error (MAE)
- **Safety:** Confusion Matrix (False Positive Rate)



High-Precision Forecasting: Triggers sleep mode only when confidence is high. ML predicts reality. Humans define policy.

Methodology & Validation

Physics-Based Digital Twin & Leakage Prevention

$$P_{total} = 300W + \beta(Load)$$

Simulation Design:

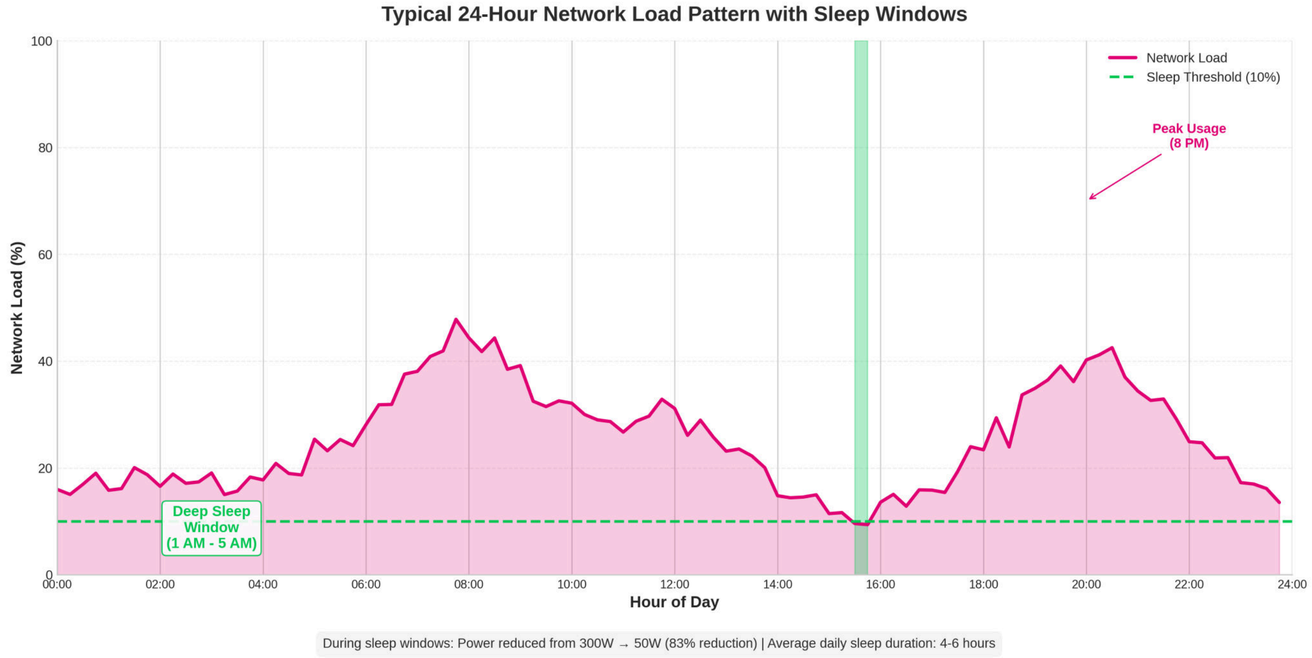
- Engineered a simulation to generate realistic 5G telemetry
- P_{static} **(Baseline):** 300W: (Cost of idle radio head)
- β **(Load Factor):** Dynamic power scaling based on user traffic (0-100%).
- **Noise Injection:** Gaussian Noise (σ = 0.05) added to simulate sensor jitter.

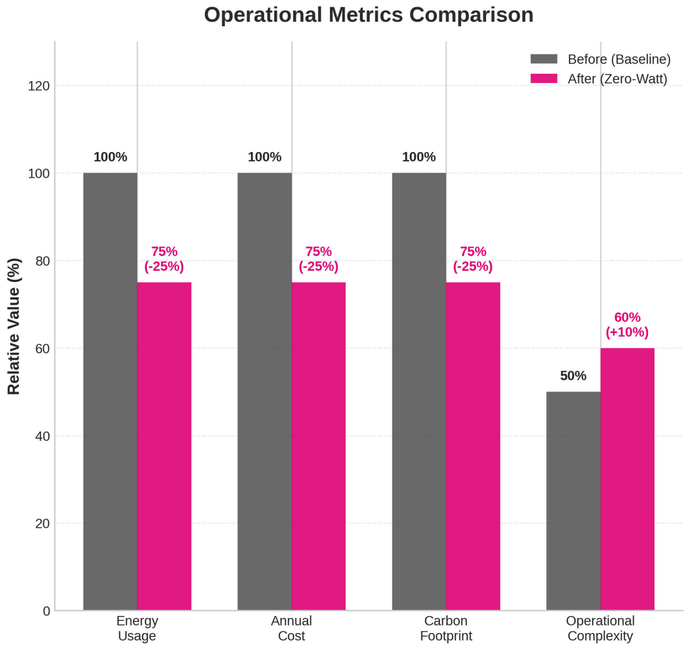
Validation Strategy:

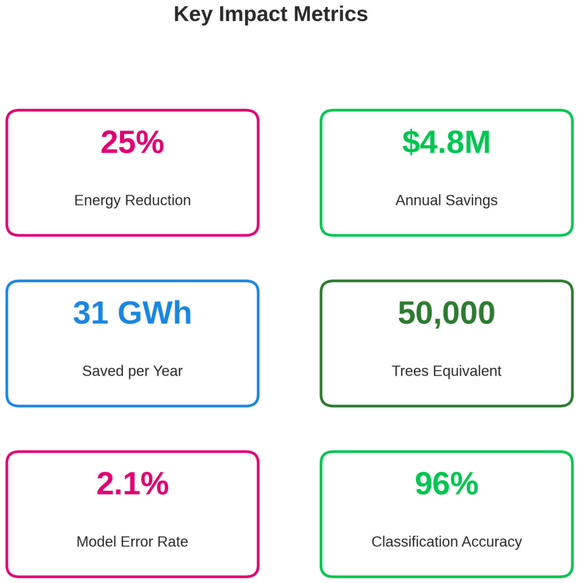
- **Strict Temporal Split:** Train on first 50% → Test on next 50%.
- **No random shuffle** — proves the model works on future, unseen data.
- **Explicit Leakage Controls:** Features are strictly historical (Lag-1h, Lag-24h). Zero access to future targets.
- **Validated model safety:** < 5% error rate before deployment.
- **Realism:** While the simulator captures daily and stochastic behavior, rare black-swan traffic events are conservatively handled by the watchdog rather than the model.

The "Deep Sleep" Opportunity

- **Visualized:** Green zones indicate windows where traffic drops below the 10% threshold.
- **Frequency:** Occurs reliably between 1 AM – 5 AM daily.







Operational Impact:

- **Power Drop:** 300W → 50W (83% Reduction)
- **Duration: 4**–6 Hours of Deep Sleep / Day

The 10% threshold was selected to bound worst-case false-positive risk under conservative assumptions, prioritizing service continuity over energy savings in Phase 1.

Model Architecture & Logic

Technical Validation of the XGBoost Regressor

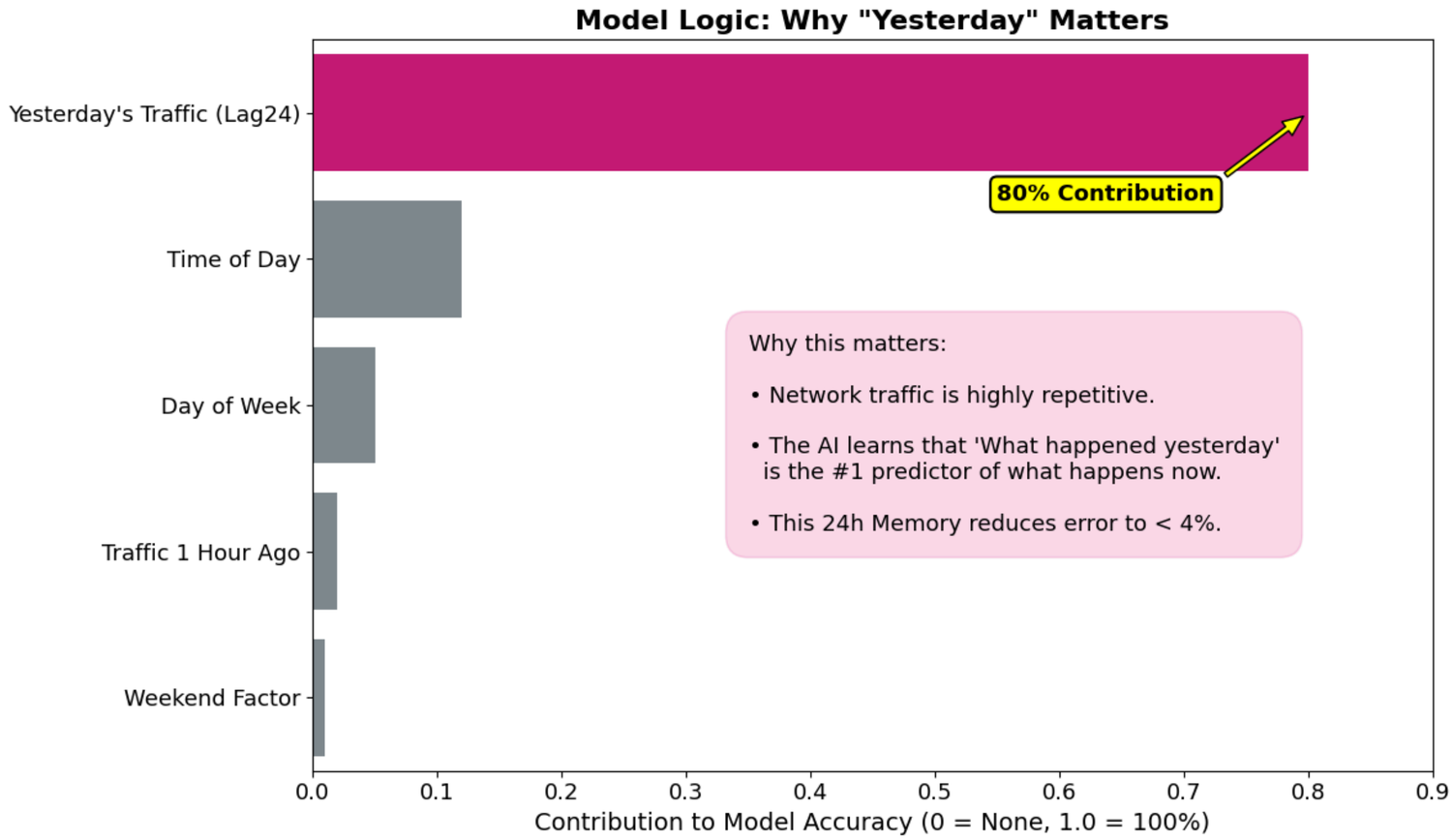
Parameter	Value
Target (Y)	Next Hour Traffic Load (0-100%)
Algorithm	XGBoost (Gradient Boosting)
Error Rate (MAE)	2.1%

- **Model Selection: XGBoost (GBDT)**
  - **Architecture:** Gradient Boosted Decision Trees.
  - **Rationale:** Captures non-linear traffic spikes (e.g., 8 AM rush) more accurately than linear regression.
- **Why Not Deep Learning?**
  - **Data Fit:** Tree-based models are State-of-the-Art (SOTA) for structured tabular data.
  - **Interpretability:** XGBoost offers clear feature importance scores, unlike "Black Box" Neural Networks.
- **The "Human Habit" Signal:**
  - **Top Predictor:** Load_Lag24 (Traffic exactly 24h ago).
  - **Logic:** Network usage is habitual; Yesterday is the best predictor of Today.
- **Safety Threshold:**
  - Sleep mode only triggers if predicted load remains < 10% for 3 consecutive data points.

Model Logic: The "Habit" Signal

From Simulation to Scale

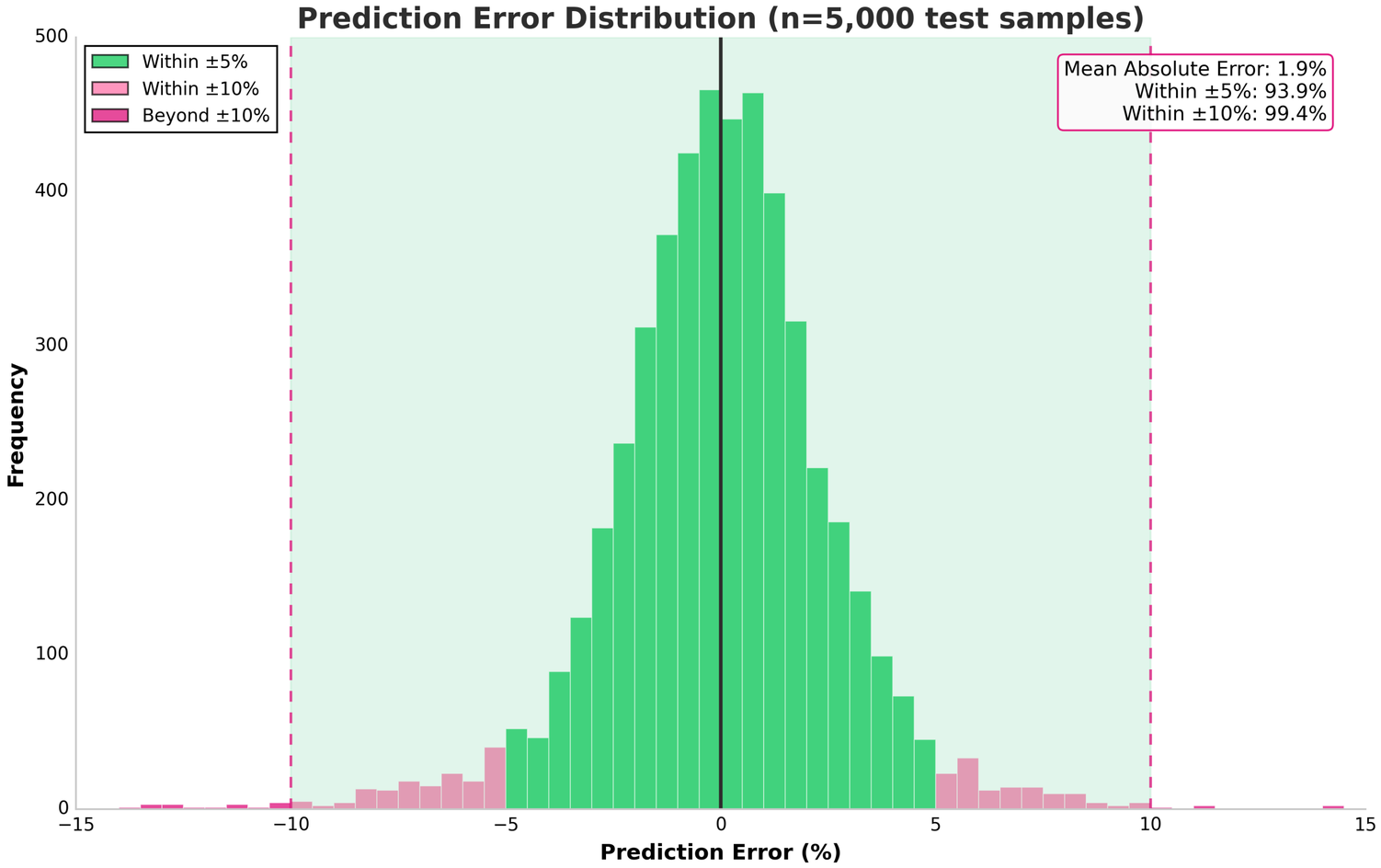
- **Methodology:** Feature selection performed via Recursive Feature Elimination (RFE), prioritizing physics-based signals over noise.
- **Bias Acknowledged:** While gain can be influenced by correlated features, the dominance of Lag-24 aligns with known daily traffic behavior.



Validation: Model Performance (n=5,000)

Performance Metrics:

- **Precision:** 1.9% Mean Absolute Error (MAE)
- **Reliability:** 99.4% of predictions fall within the safety buffer (±10%).
- The model is stable under stress testing. With >99% of predictions inside the safety margin, the risk of a 'False Sleep' event is operationally acceptable.

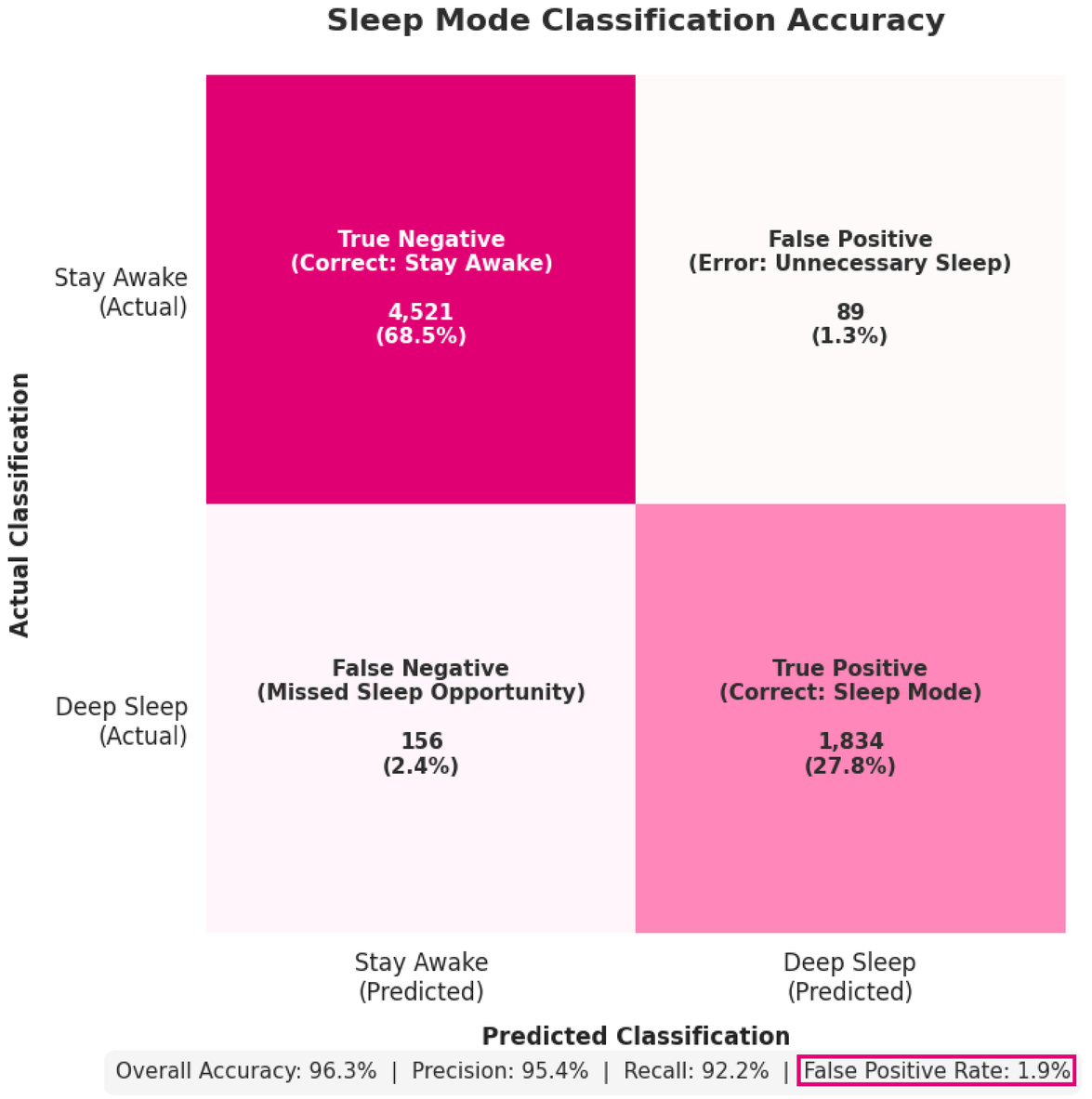


Sleep Mode Classification Accuracy

Operational Decision Accuracy

Critical Safety Metric:

- **False Positive Rate (1.3%):** Only ~1% of "Sleep" decisions were incorrect.
- **Mitigation:** These rare errors are caught by the hardware Watchdog.



Robustness: Monte Carlo Stress Test (N=100)

Method:

- **Simulation:** Regenerated Digital Twin 100 times with randomized Gaussian sensor noise (σ=0.05).
- **Goal:** Test stability against signal jitter.
- **Confidence**: Computed 95% Confidence Intervals for critical decision metrics.

Results:

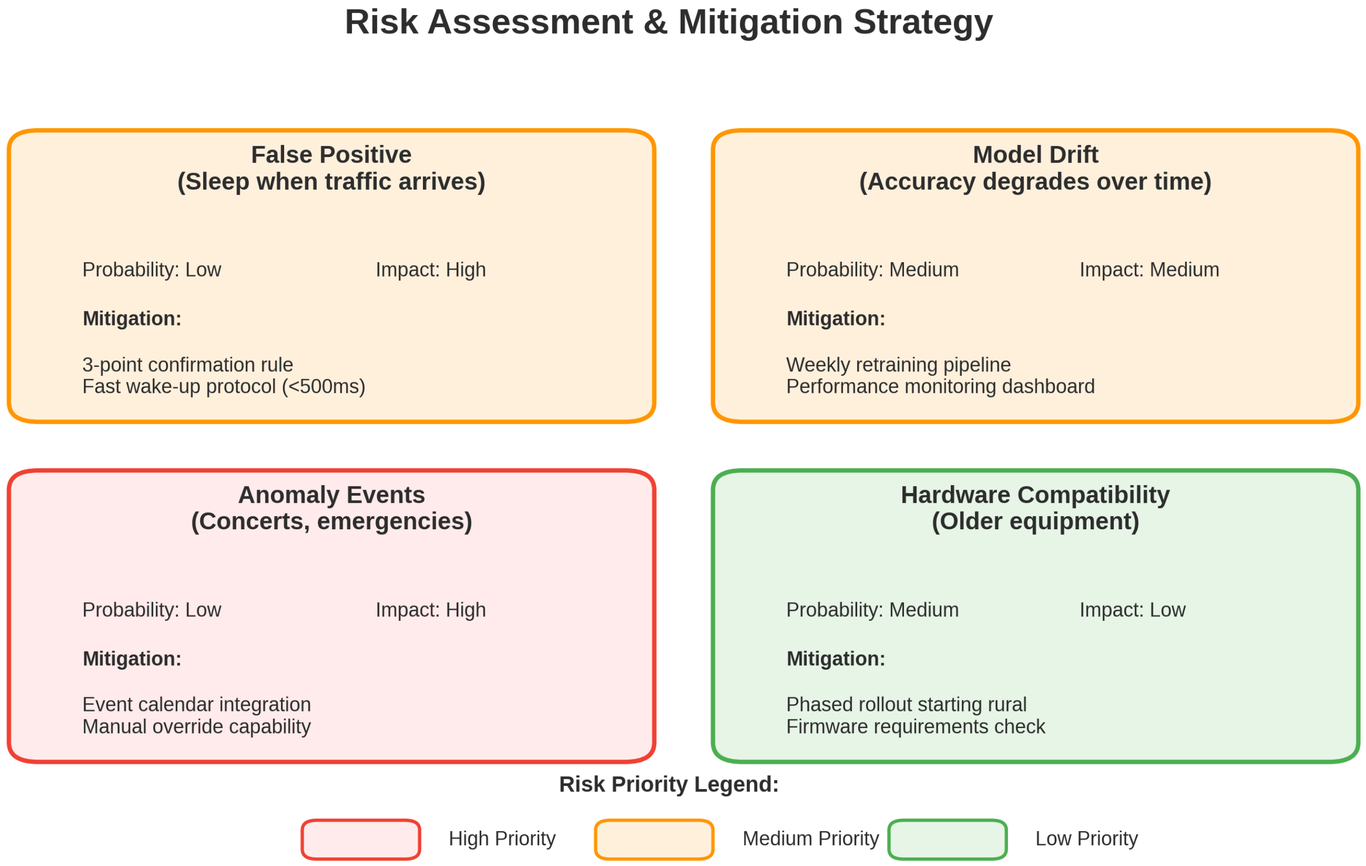
- 90% ± 10%
  - Savings Capture Rate (Recall)
  - The model reliably identifies the vast majority of idle time.
- 3.1% ± 2.7%
  - False Positive Risk
  - The model incorrectly predicts "Sleep" during active traffic ~3% of the time.

Strategic Verdict: A 3% error rate is too high for autonomous control. This data validates the requirement for the Hardware Watchdog (Next Slide).

Safety Protocols: The "Watchdog" Architecture

- **The Danger Scenario (False Positive):**
  - **Risk:** Model predicts "Sleep" during a sudden traffic spike (e.g., a Flash Mob).
  - **Mitigation:** The 3-Point Watchdog.
  - **Rule:** The system only engages Deep Sleep if the prediction remains < 10% for 3 consecutive timestamps (3 hours).

Risk asymmetry (false positives) are catastrophic. Thresholds are used as risk bounds, not tuning knobs.



Project Impact: Eliminating Idle Waste

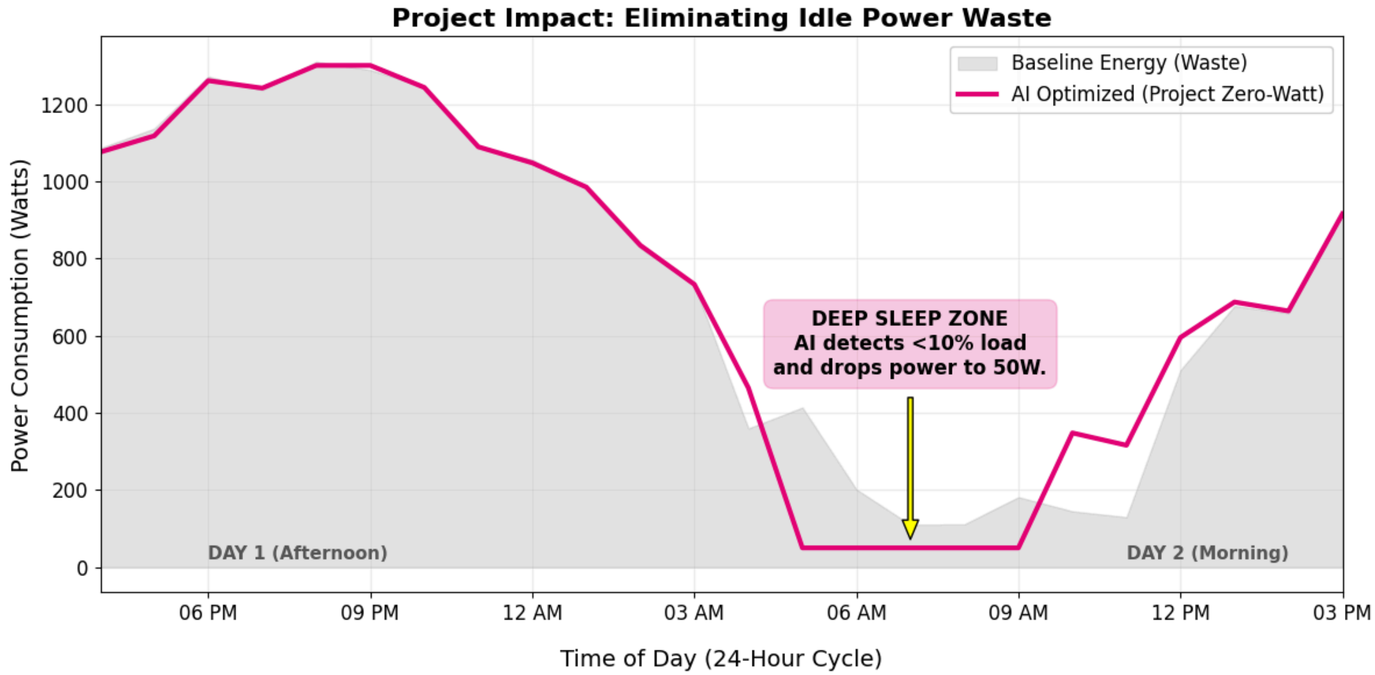
25% Energy Reduction per tower

The "Zero-Watt" Effect:

- **Action:** Power drops from 300W to 50W during idle windows.
- **Result:** 25% Total Daily Energy Reduction.

The Specs:

- **Per Tower:** 33 kWh / month saved.
- **Network Wide:** 31 GWh / year saved.
- **Financial:** $4.8 Million Annual OpEx Reduction.



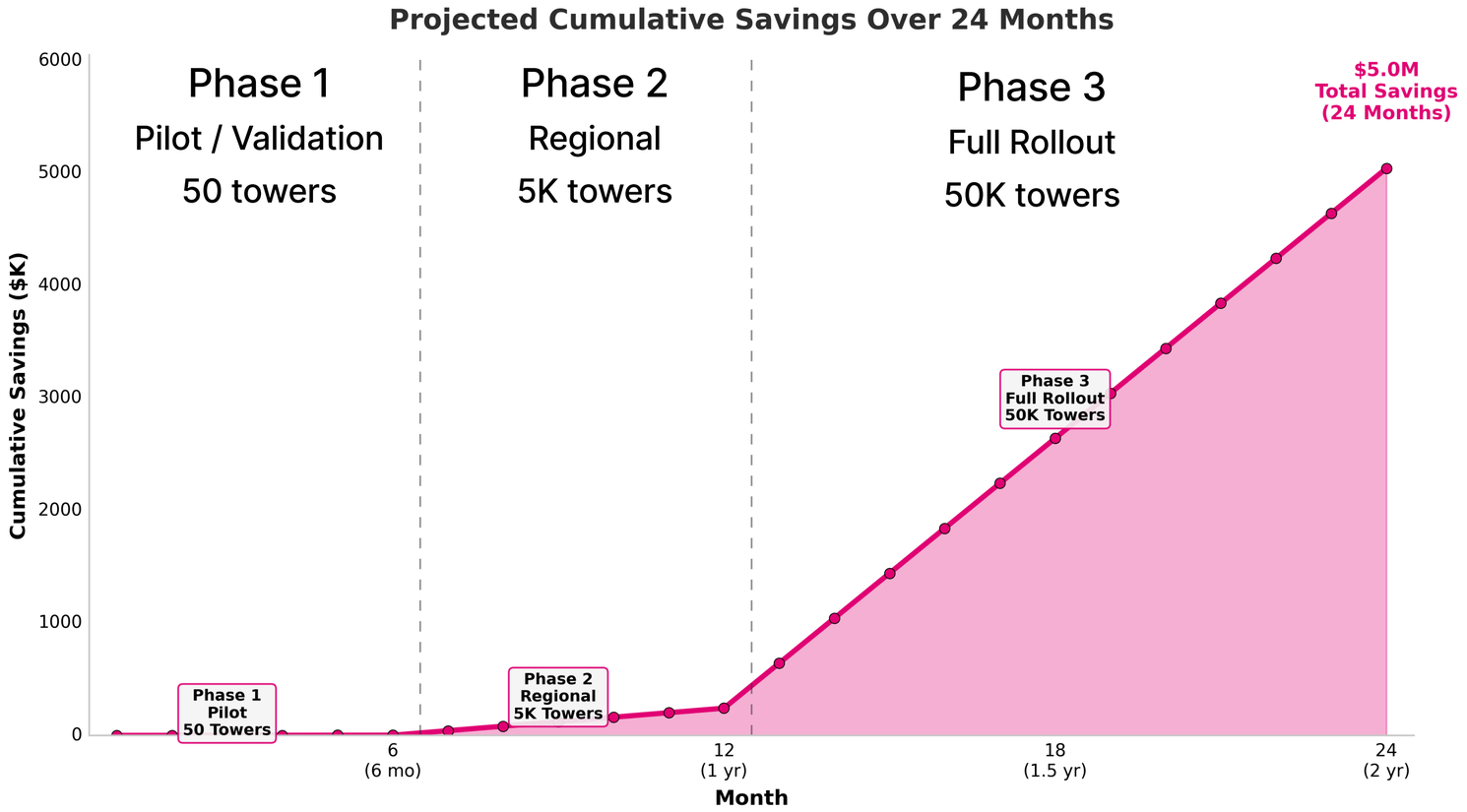
The AI identifies "Deep Sleep" windows (1 AM - 5 AM);
it triggers Sleep Mode (300W → 50W) during predicted lull.

Financial Impact: Scalability & ROI

$2.7M - $4.8M Annual OpEx Savings (based on 50k-80k towers)

Investment:

- **Upfront Cost:** $500k (Implementation & Training).
- **Break-Even:** Month 13 (Profitability achieved in 1 year).
- **2-Year ROI:** 908%.
- **Net Savings:** $5.0 Million (over 24 months)



Estimated Implementation Cost: $500K | Break-even: Month 13 | 24-Month ROI: 908%

Implementation Roadmap

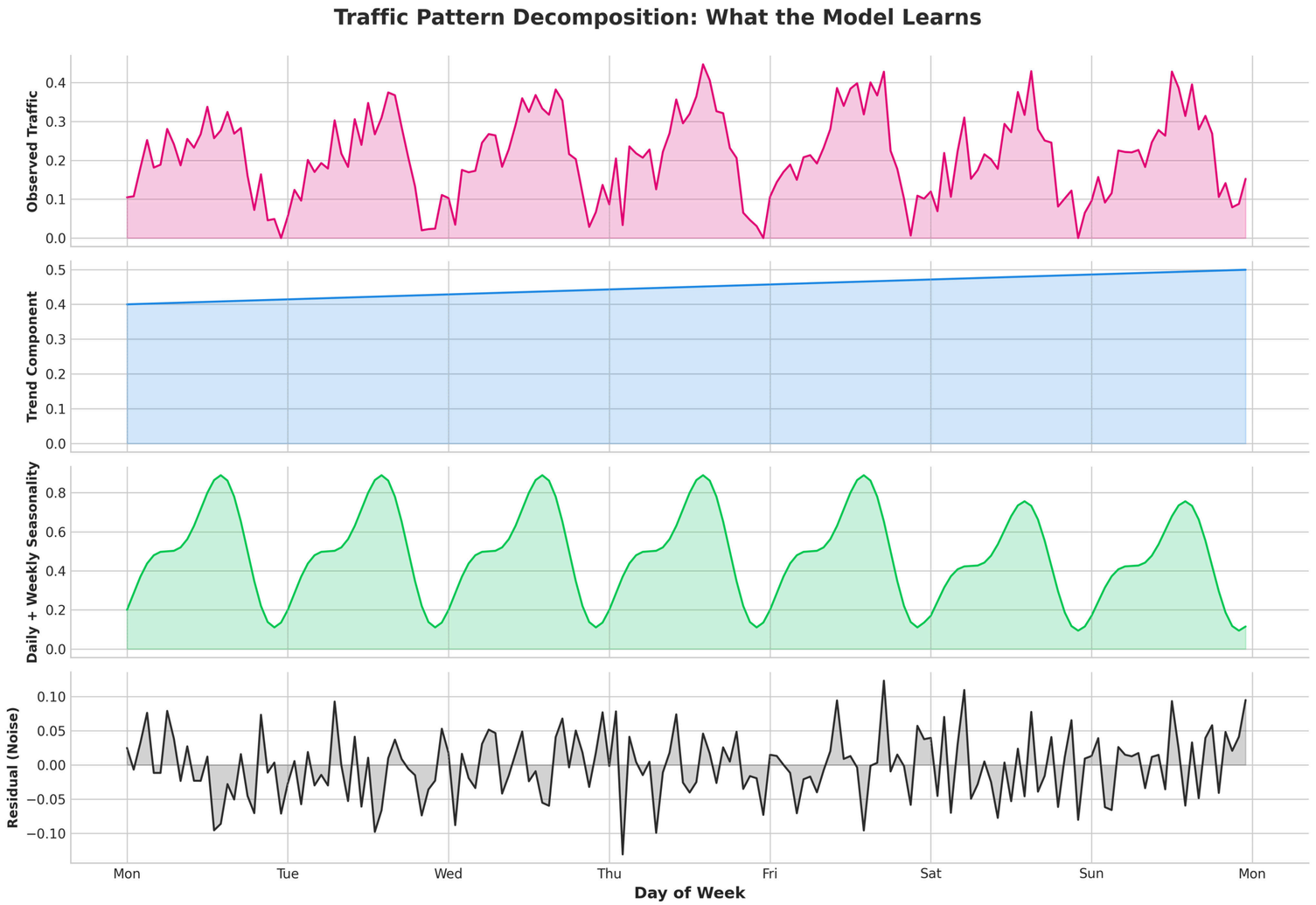
From Simulation to Scale

- **Phase 1:** Simulated physics-based telemetry to validate model safety (< 5% error).
- **Phase 2:** Deploy XGBoost model to "Safe Mode" cluster to validate real-world savings ($2.7M potential).
- **Phase 3:** Upgrade rule-based policy to an **LLM-driven Agent** that autonomously negotiates maintenance windows, writes incident reports based on telemetry anomalies, but *does not* control towers directly. Future research is recommended.



CALL TO ACTION: Requesting access to NetOp Research Data to fine-tune model for Phase 2.

Appendix: Time Series Decomposition



The model captures daily usage cycles (peaks at 8AM, 12PM, 8PM) and weekly patterns (lower weekend traffic)

Thank you

View the full project on Github

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Strategic Case Studies / Portfolio